

Quarter 1 Assessment Blueprint

Benchmark	MA.7.NSO.1.2	MA.7.AR.2.2	MA.8.AR.2.2	MA.8.AR.2.3	MA.8.NSO.1.1	MA.8.NSO.1.2
# of Items	4	4	4	3	3	3
Unit of Instruction	Unit 1	Unit 1	Unit 1	Unit 2	Unit 2	Unit 2
2024-2025 District Percent Correct	51%	52%	40%	35%	65%	47%
Benchmark	MA.7.NSO.1.1	MA.8.NSO.1.3	MA.8.NSO.1.4			
# of Items	3	3	3			
Unit of Instruction	Unit 3	Unit 3	Unit 3			
2024-2025 District Percent Correct	46%	49%	63%			

Total Number of Items: 30

Quarter 1
Recommended Pacing Calendar

August '25						
Su	M	Tu	W	Th	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

September '25						
Su	M	Tu	W	Th	F	S
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

October '25						
Su	M	Tu	W	Th	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1

Quarter 1 Summary of Instructional Units

Units are linked to the SCPS Frameworks site. Bolded benchmarks are assessed on the current quarter assessment. The benchmarks that are not in bold may be assessed in the following quarter.

Unit 1 - Equations and Inequalities	With 10 lessons over 12 recommended instructional days, this unit covers benchmarks MA.7.NSO.1.2, MA.7.AR.2.2, and MA.8.AR.2.2. In this unit, students engage in writing and solving two-step equations and inequalities in mathematical and real-world contexts. This unit picks up from the end of Grade 6 Accelerated Math where students learned to write and solve one-step equations and inequalities.
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	In Section 1, students are introduced to repeating decimals and learn how to convert from fractions to repeating decimals and from repeating decimals to fractions. This builds on students’ prior learning of how to rewrite rational numbers in different but equivalent forms from Grade 6 Accelerated Math. Section 2 extends students’ prior knowledge of solving one-step equations as they explore solving two-step equations using the properties of equality. Students write and solve two-step equations that represent real-world contexts. Students informally check their solutions by considering if they are reasonable in the given context. They also formally check their solutions using substitution. Section 3 focuses on inequalities. The section starts by reintroducing inequalities and having students determine which solutions will make the inequality true. Students then go on to apply what they know about solving two-step equations to solve two-step inequalities, ultimately being able to solve and graph the solutions to a two-step inequality.
Unit 2 - Rational and Irrational Numbers	With 6 lessons over 8 recommended instructional days, this unit covers benchmarks MA.8.AR.2.3, MA.8.NSO.1.1, and MA.8.NSO.1.2. In this unit, students develop an understanding of irrational numbers as they continue their exploration of the real number system. In Section 1, students explore the geometric representation, algebraic notation, and verbal descriptions of perfect squares and square roots. Then they explore the geometric representations, algebraic notation, and verbal descriptions of perfect cubes and cube roots. Students solve equations in the form $x^2=p$ and $x^3=q$, where p is a whole number and q is an integer. They also learn to find the values of square roots and cube roots. In Section 2, students extend their knowledge of rational numbers to define and explore irrational numbers. Students use their knowledge of finding the square and cube roots of perfect squares and cubes to approximate non-perfect square and cube roots. They finish the unit by plotting, comparing, and ordering rational numbers and the approximated values of irrational numbers on a number line.
Unit 3 - Exponents and Scientific Notation	With 9 lessons over 11 recommended instructional days, this unit covers benchmarks MA.7.NSO.1.1, MA.8.NSO.1.3, and MA.8.NSO.1.4. In this unit, students are introduced to the laws of exponents and learn to use them to evaluate numeric expressions. They are also introduced to scientific notation. Section 1 focuses on the laws of exponents, their definitions, and their applications with whole number exponents. In this section, the laws of exponents are broken into two categories: product exponent laws and quotient exponent laws. Students conclude this section by evaluating numeric expressions and generating equivalent expressions by applying the laws of exponents. In Section 2, students extend the laws of exponents to include integer exponents as they explore negative exponents. Students apply the laws of exponents to evaluate numerical expressions in addition to generating equivalent expressions. In Section 3, students explore how to write large and small numbers in scientific notation with powers of 10. Students discover how to rewrite numbers expressed in standard form in scientific notation and vice versa. Students apply the laws of exponents to integer powers of 10 and determine how many times greater or smaller a number written in scientific notation is than another number written in scientific notation.
Unit 4 - Solving Problems with Rational Numbers	With 8 lessons over 10 recommended instructional days, this unit covers benchmarks MA.8.NSO.1.5, MA.8.NSO.1.6, and MA.8.NSO.1.7 <i>While instruction on this unit takes place at the end of Quarter 1, this unit will not be assessed on the Quarter 1 Benchmark Assessment.</i> In Unit 4, students build on prior knowledge from Grade 6 Accelerated and from Unit 3 to perform operations with numbers expressed in scientific notation and evaluate expressions using the order of operations. In Section 1, students explore how to perform operations on numbers written in scientific notation in real-world and mathematical problems. Through guided instruction, practice, error analysis tasks, and collaborative activities, students learn how to add, subtract, multiply, and divide numbers written in scientific notation.

In Section 2, students build on their understanding of evaluating expressions with rational numbers, grouping symbols, and exponents to extend to include expressions with radicals. Students experience scaffolded learning through solving mathematical problems involving the order of operations to understand how a value is affected when operations are performed in a different order.